

of this was that it allowed for the placing of accurate clocks in space, a technology that GPS heavily relies on.

GPS was developed as NAVSTAR (navigation system with timing and ranging). Before its civilian applications, GPS was used to provide all-weather round-the-clock navigation capabilities for military ground, sea and air forces.

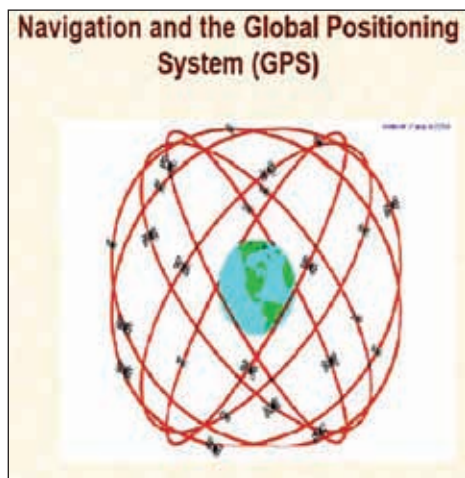
GPS now has applications beyond navigation and location determination. GPS can be used for geography, forestry, mineral exploration, wildlife habitation management, monitoring the movement of people and things and bringing precise timing to the world. It was recently used to re-measure the height of Mt Everest.

GPS satellites continuously transmit digital radio signals that contain data on the satellites' location and the exact time to earth-bound receivers. The satellites are equipped with atomic clocks that are accurate to within one-billionth of a second. Based on the information received, the receivers calculate how long it takes for the signal to reach them, and thus how far away each satellite is.

Initially, the GPS system was intended only for military uses and GPS signals available to civilians were initially distorted. This gave errors of up to 100 metres at times. The resulting

average error due to this was 10 metres (32 ft) horizontally and 30 metres (98 ft) vertically.

This process of distorting the accuracy of civilian GPS signals was called selective availability. Selective availability was intended to deny an enemy the use of civilian



A diagram of the satellites in the GPS system